

Isaac S. Garee

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EDUCATION

University of Cincinnati

Expected May 2027

Bachelor of Science, Industrial and Systems Engineering

GPA: 3.5

Minors, Mathematics and Statistics

- Primary Coursework: Linear Algebra, Deterministic Systems Models, Probabilistic Systems Models, Data Science, Python Programming, Probability and Statistics

EXPERIENCE

University of Southern California | Los Angeles, California

May 2026 – Aug 2026

Data Science and Operations Summer Scholar, Advised by Prof. Andrew Daw and Vishal Gupta

- Developed stochastic simulation models in Python to evaluate inventory, throughput, stockout risk, and manufacturing flexibility in multi-component assembly systems
- Designed and executed computational experiments across broad parameter spaces to identify system bottlenecks, quantify performance tradeoffs, and develop data-driven recommendations
- Extended a single-product supply chain model to multi-product systems with shared and product-specific components, integrating stochastic simulation with mathematical analysis to study resilient manufacturing operations
- Presented project status, technical risks, and preliminary findings through weekly meetings and a final presentation.

University of Cincinnati | Cincinnati, Ohio

Apr 2025 – May 2026

Undergraduate Researcher, Advised by Prof. Tommaso Giovannelli

- Engineered a novel mixed-integer bilevel optimization Neural Architecture Search (NAS) framework, delivering a 47% speedup over the current baselines while maintaining competitive accuracy
- Designed and implemented a PyTorch codebase from scratch including finite-difference gradients, Hessian-free updates, and line-search functions, enabling scalable experimentation across architecture search spaces
- Benchmarked algorithm convergence and stability against existing NAS baselines, evaluating architecture quality, training dynamics, and validation accuracy
- Proved key theoretical properties of the mixed-integer bilevel algorithm in collaboration with faculty advisor, contributing to a research manuscript currently in preparation

Kaivac | Hamilton, Ohio

Jan 2024 – Dec 2024

Industrial and Systems Engineering Co-op

- Reduced inbound inspection process time by 20% by identifying bottlenecks through value stream mapping and implementing targeted process improvements
- Delivered Power BI dashboards that unified cross-functional production data, enabling faster management decisions and eliminating manual reporting workflows
- Deployed control charts and SPC metrics across manufacturing lines, introducing real-time quality monitoring where none previously existed
- Expanded data tracking infrastructure to capture previously unmeasured process variables, generating new insights that sustained continuous improvement efforts

SKILLS

Programming and Tools: Python, R, PyTorch, NumPy, SciPy, Pandas, LaTeX, Gurobi, Excel, Power BI

Technical Skills: Bilevel Optimization, Machine Learning, Value Stream Mapping, Stochastic Modeling

ACTIVITIES & HONORS

Industrial & Systems Engineering Student Association | Secretary

Jan 2024 – Jan 2025

- Coordinated meeting logistics, recorded minutes, and organized guest speaker sessions for a 50+ member engineering organization

Shidecker Memorial Golden Eagle Scholarship

Aug 2022 – May 2023

- Awarded a scholarship from Shidecker Memorial Foundation in recognition of commitment to both academic excellence and community involvement